

Sequency Decomposition of Jacobian Matrix

A brief discussion about sequencies is in order to start. The entire hierarchy is based on octaves of 50% duty cycle square waves. These are essentially the signum function of harmonics of a sine wave, or an infinitely amplified sine wave, clipped at a unit magnitude. Although it is clearly a digital waveform, it is derived as the limit of an analog waveform. In this respect it can be applied in much the same manner as harmonics in a Fourier transformation. But where a Fourier transformation requires 4 quadrant analog multiplication and integration, the digital nature of a square wave reduces that to addition and subtraction plus summation. With sine wave harmonics, each zero-crossing is uniformly spaced. This is not the case with sequencies. but, in exchange for equal spacing, all zero crossings fall on top of one another. This facilitates a parallel processing algorithm. Since the dominant sine wave component in any sequency is the same as half the count of zero-crossings in one whole period of the fundamental, a good sense of the Fourier spectrum of any arbitrary waveform can be obtained much faster by sequency decomposition.

The primary sequencies are known as Rademacher functions. They are all octaves above the fundamental. The spaces inbetween are the result of products of Rademacher functions, up to and including all of a set. As a result, every sequency from 0 to $2^n - 1$ can be synthesized by some combination of harmonics. One of the features of square waves is that the product of any two square waves (primary or not) is another square wave in the set. Sine waves have the same frequency shifting as a result of multiplication, but the product waveform is not a sine wave. Square waves form a closed group with respect to multiplication. Since this is actually how square waves are defined, there is another shortcut to multiplication. Because the product of any square wave with itself is the 0 sequency, there can never be more than 1 copy of any factor. So, regardless of composition, the sequency of the product of any two factors is just the XOR of the sequency numbers of the two factors. The product of any two sequencies can thus be either higher

or lower in sequency than either factor. But it means that sequencies with 32000 steps can be multiplied as easily as sequencies with 2 or 4 steps.

Unlike the hypercomplex numbers, with which they have some common properties, sequencies are unlimited and retain both commutativity and associativity. Here is a chart of the first few sets:

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The 2x2 matrix is represented by the transformation from Minkowski spacetime coordinates to eigenspace (plus a little scaling, which is indeterminate for eigenvectors). The 4x4 matrix represents the transformation from 1/3 Cartesian coordinates to 4 tetrahedral coordinates. The 8x8 matrix transforms the corners of a cube into a combination of Cartesian and degenerate eigenspace vectors. The 16x16 matrix is the focus of this presentation. Given a complex 4-differential operator and a complex 4-potential, the product is a 16 element set. By a combination of conjugates of 4 different types, a set of 16 conjugates of the complex product results. It just seemed like an invitation to test sequency decomposition. The results were not disappointing.

To create a sequency decomposition matrix from this listing of Walsh functions, we only need to remember that the product of this matrix with itself is the identity matrix. For each row|column pair, that means the product is either 0 or 1. But there are 2^n contributions for each nxn matrix. To normalize the result, depending on the size of the matrix, we must divide each unit by $\sqrt[4]{(2^n)}$. That way, each product is $1/2^n$, and the sum of 2^n contributions is just 1.

$$\sqrt{2} \begin{vmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{vmatrix} \begin{vmatrix} ct \\ r \end{vmatrix} = \begin{vmatrix} \Sigma \\ \Delta \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} \begin{vmatrix} ct \\ r \end{vmatrix}$$

$$\frac{1}{\sqrt{2}} \begin{vmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{vmatrix} \begin{vmatrix} \Sigma \\ \Delta \end{vmatrix} = \begin{vmatrix} ct \\ r \end{vmatrix} = \begin{vmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{vmatrix} \begin{vmatrix} \Sigma \\ \Delta \end{vmatrix}$$

$$\begin{vmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{vmatrix} \begin{vmatrix} (+1,+1,+1) \\ (+1,-1,-1) \\ (-1,-1,+1) \\ (-1,+1,-1) \end{vmatrix} = \begin{vmatrix} (0,0,0) \\ (2,0,0) \\ (0,2,0) \\ (0,0,2) \end{vmatrix}$$

$$\begin{vmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{vmatrix} \begin{vmatrix} (0,0,0) \\ (2,0,0) \\ (0,2,0) \\ (0,0,2) \end{vmatrix} = \begin{vmatrix} (+1,+1,+1) \\ (+1,-1,-1) \\ (-1,-1,+1) \\ (-1,+1,-1) \end{vmatrix}$$

$$\frac{1}{\sqrt{8}} \begin{vmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\ 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 \\ 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \\ 1 & -1 & -1 & 1 & -1 & 1 & 1 & -1 \\ 1 & -1 & 1 & -1 & -1 & 1 & -1 & 1 \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \end{vmatrix} \begin{vmatrix} (+1, +1, +1) \\ (+1, +1, -1) \\ (+1, -1, -1) \\ (+1, -1, +1) \\ (-1, -1, +1) \\ (-1, -1, -1) \\ (-1, +1, -1) \\ (-1, +1, +1) \end{vmatrix} = \begin{vmatrix} (0, 0, 0) \\ (\sqrt{8}, 0, 0) \\ (0, \sqrt{8}, 0) \\ (0, 0, 0) \\ (0, 0, \sqrt{8}) \\ (0, 0, 0) \\ (0, 0, 0) \\ (0, 0, 0) \end{vmatrix}$$

$$\frac{1}{\sqrt{8}} \begin{vmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\ 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 \\ 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \\ 1 & -1 & -1 & 1 & -1 & 1 & 1 & -1 \\ 1 & -1 & 1 & -1 & -1 & 1 & -1 & 1 \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \end{vmatrix} \begin{vmatrix} (0, 0, 0) \\ (\sqrt{8}, 0, 0) \\ (0, \sqrt{8}, 0) \\ (0, 0, 0) \\ (0, 0, \sqrt{8}) \\ (0, 0, 0) \\ (0, 0, 0) \\ (0, 0, 0) \end{vmatrix} = \begin{vmatrix} (+1, +1, +1) \\ (+1, +1, -1) \\ (+1, -1, -1) \\ (+1, -1, +1) \\ (-1, -1, +1) \\ (-1, -1, -1) \\ (-1, +1, -1) \\ (-1, +1, +1) \end{vmatrix}$$

Definitions:

$$\Diamond = \partial/c\partial t \ l + \partial/\partial z \ i + \partial/\partial x \ j + \partial/\partial y \ k$$

$$\Diamond^* = \partial/c\partial t \ l - \partial/\partial z \ i - \partial/\partial x \ j - \partial/\partial y \ k$$

$$A = \varphi \ l + A_z \ i + A_x \ j + A_y \ k$$

$$A^* = \varphi \ l - A_z \ i - A_x \ j - A_y \ k$$

$$[\Diamond A] = [\Diamond]^t [A] =$$

$$\begin{vmatrix} \partial\varphi/c\partial t \ l & \partial A_z/c\partial t \ i & \partial A_x/c\partial t \ j & \partial A_y/c\partial t \ k \\ \partial\varphi/\partial z & i & \partial A_z/\partial z & -l & \partial A_x/\partial z & k & \partial A_y/\partial z & -j \\ \partial\varphi/\partial x & j & \partial A_z/\partial x & -k & \partial A_x/\partial x & -l & \partial A_y/\partial x & i \\ \partial\varphi/\partial y & k & \partial A_z/\partial y & j & \partial A_x/\partial y & -i & \partial A_y/\partial y & -l \end{vmatrix}$$

Conjugates only change the sign of the element in the product matrix:

$$\begin{matrix} \Diamond A & \Diamond A^* & \Diamond^* A & \Diamond^* A^* \\ \begin{vmatrix} + & + & + & + \\ + & + & + & + \\ + & + & + & + \\ + & + & + & + \end{vmatrix} & \begin{vmatrix} + & - & - & - \\ + & - & - & - \\ + & - & - & - \\ + & - & - & - \end{vmatrix} & \begin{vmatrix} + & + & + & + \\ - & - & - & - \\ - & - & - & - \\ - & - & - & - \end{vmatrix} & \begin{vmatrix} + & - & - & - \\ - & + & + & + \\ - & + & + & + \\ - & + & + & + \end{vmatrix} \end{matrix}$$

$$\begin{array}{cccc}
-\Diamond A & -(\Diamond A^*) & -(\Diamond^* A) & -(\Diamond^* A^*) \\
\left| \begin{array}{cccc} - & - & - & - \\ - & - & - & - \\ - & - & - & - \\ - & - & - & - \end{array} \right| & \left| \begin{array}{cccc} - & + & + & + \\ - & + & + & + \\ - & + & + & + \\ - & + & + & + \end{array} \right| & \left| \begin{array}{cccc} - & - & - & - \\ + & + & + & + \\ + & + & + & + \\ + & + & + & + \end{array} \right| & \left| \begin{array}{cccc} - & + & + & + \\ + & - & - & - \\ + & - & - & - \\ + & - & - & - \end{array} \right| \\
-(\Diamond A)^* & -(\Diamond A^*)^* & -(\Diamond^* A)^* & -(\Diamond^* A^*)^* \\
\left| \begin{array}{cccc} - & + & + & + \\ + & - & + & + \\ + & + & - & + \\ + & + & + & - \end{array} \right| & \left| \begin{array}{cccc} - & - & - & - \\ + & + & - & - \\ + & - & + & - \\ + & - & - & + \end{array} \right| & \left| \begin{array}{cccc} - & + & + & + \\ - & + & - & - \\ - & - & + & - \\ - & - & - & + \end{array} \right| & \left| \begin{array}{cccc} - & - & - & - \\ - & - & + & + \\ - & + & - & + \\ - & + & + & - \end{array} \right|
\end{array}$$

| | | | |
|--|--|--|--|
| $[\Diamond A]$ | $[\Diamond A^*]$ | $[\Diamond^* A]$ | $[\Diamond^* A^*]$ |
| $\begin{vmatrix} + & + & + & + \\ + & - & + & - \\ + & - & - & + \\ + & + & - & - \end{vmatrix}$ | $\begin{vmatrix} + & - & - & - \\ + & + & - & + \\ + & + & + & - \\ + & - & + & + \end{vmatrix}$ | $\begin{vmatrix} + & + & + & + \\ - & + & - & + \\ - & + & + & - \\ - & - & + & + \end{vmatrix}$ | $\begin{vmatrix} + & - & - & - \\ - & - & + & - \\ - & - & - & + \\ - & + & - & - \end{vmatrix}$ |
| $[(\Diamond A)^*]$ | $[(\Diamond A^*)^*]$ | $[(\Diamond^* A)^*]$ | $[(\Diamond^* A^*)^*]$ |
| $\begin{vmatrix} + & - & - & - \\ - & - & - & + \\ - & + & - & - \\ - & - & + & - \end{vmatrix}$ | $\begin{vmatrix} + & + & + & + \\ - & + & + & - \\ - & - & + & + \\ - & + & - & + \end{vmatrix}$ | $\begin{vmatrix} + & - & - & - \\ + & + & + & - \\ + & - & + & + \\ + & + & - & + \end{vmatrix}$ | $\begin{vmatrix} + & + & + & + \\ + & - & - & + \\ + & + & - & - \\ + & - & + & - \end{vmatrix}$ |
| $[-\Diamond A]$ | $[-(\Diamond A^*)]$ | $[-(\Diamond^* A)]$ | $[-(\Diamond^* A^*)]$ |
| $\begin{vmatrix} - & - & - & - \\ - & + & - & + \\ - & + & + & - \\ - & - & + & + \end{vmatrix}$ | $\begin{vmatrix} - & + & + & + \\ - & - & + & - \\ - & - & - & + \\ - & + & - & - \end{vmatrix}$ | $\begin{vmatrix} - & - & - & - \\ + & - & + & - \\ + & - & - & + \\ + & + & - & - \end{vmatrix}$ | $\begin{vmatrix} - & + & + & + \\ + & + & - & + \\ + & + & + & - \\ + & - & + & + \end{vmatrix}$ |
| $[-(\Diamond A)^*]$ | $[-(\Diamond A^*)^*]$ | $[-(\Diamond^* A)^*]$ | $[-(\Diamond^* A^*)^*]$ |
| $\begin{vmatrix} - & + & + & + \\ + & + & + & - \\ + & - & + & + \\ + & + & - & + \end{vmatrix}$ | $\begin{vmatrix} - & - & - & - \\ + & - & - & + \\ + & + & - & - \\ + & - & + & - \end{vmatrix}$ | $\begin{vmatrix} - & + & + & + \\ - & - & - & + \\ - & + & - & - \\ - & - & + & - \end{vmatrix}$ | $\begin{vmatrix} - & - & - & - \\ - & + & + & - \\ - & - & + & + \\ - & + & - & + \end{vmatrix}$ |

To perform the decomposition, we lay out all the signs in linear order:

| | | |
|-------------------------|---|-----------------------------------|
| $[(\Diamond A)]$ | = | [+ + + + + - + - + - - + + + - -] |
| $[(\Diamond A^*)]$ | = | [+ - - - + + - + + + + - + - + +] |
| $[(\Diamond^* A)]$ | = | [+ + + + - + - + - + + - - - + +] |
| $[(\Diamond^* A^*)]$ | = | [+ - - - - - + - - - - + - + - -] |
| $[(\Diamond A)^*]$ | = | [+ - - - - - - + - + - - - + - -] |
| $[(\Diamond A^*)^*]$ | = | [+ + + + - + + - - - + + - + - +] |
| $[(\Diamond^* A)^*]$ | = | [+ - - - + + + - + - + + + + - +] |
| $[(\Diamond^* A^*)^*]$ | = | [+ + + + + - - + + + - - + - + -] |
| $[-(\Diamond A)]$ | = | [- - - - - + - + - + + + - - + +] |
| $[-(\Diamond A^*)]$ | = | [- + + + - - + - - - - + - + - -] |
| $[-(\Diamond^* A)]$ | = | [- - - - + - + - + - - + + + - -] |
| $[-(\Diamond^* A^*)]$ | = | [- + + + + + - + + + + - + - + +] |
| $[-(\Diamond A)^*]$ | = | [- + + + + + + - + - + + + + - +] |
| $[-(\Diamond A^*)^*]$ | = | [- - - - + - - + + + - - + - + -] |
| $[-(\Diamond^* A)^*]$ | = | [- + + + - - - + - + - - - - + -] |
| $[-(\Diamond^* A^*)^*]$ | = | [- - - - - + + - - - + + - + - +] |

A quick check confirms that each of the 16 columns represents a pure sequency. Consequently when we perform the sequency decomposition, the sum of each column is only non-zero for the same sequency as the pattern. All we have to do is count the number of polarity inversions and attach the sign of the first row (no conjugates) to the amplitude. Technically, each element in the sequency decomposition matrix should be $\pm\frac{1}{4}$. Then, the result of a summation should be $16/4 = 4$ times the amplitude of the differential term. However, real space and sequency space are separated by a rotation angle, and the cosine projection of every vector to every other one is $\frac{1}{4}$. This is so whether we project 1 at a time or all at once. So, each spectral amplitude gets scaled back down by the same projection cosine, and the results are the traditional vector derivatives.

Jacobian Matrix

Signs and sequencies:

[+1 +D +D +D +5 -9 +A -A +5 -A -9 +A +5 +A -A -9]

Sequency Decomposition of the Jacobian Matrix:

$$\begin{vmatrix} \partial\phi/c\partial t & lW1 & \partial A_z/c\partial t & iWD & \partial A_x/c\partial t & jWD & \partial A_y/c\partial t & kWD \\ \partial\phi/\partial z & iW5 & -\partial A_z/\partial z & lW9 & \partial A_x/\partial z & kWA & -\partial A_y/\partial z & jWA \\ \partial\phi/\partial x & jW5 & -\partial A_z/\partial x & kWA & -\partial A_x/\partial x & lW9 & \partial A_y/\partial x & iWA \\ \partial\phi/\partial y & kW5 & \partial A_z/\partial y & jWA & -\partial A_x/\partial y & iWA & -\partial A_y/\partial y & lW9 \end{vmatrix}$$

$$\begin{aligned} & +(\partial\phi/c\partial t \quad l) \quad W1 \\ & +(\partial\phi/\partial z \quad i) + (\partial\phi/\partial x \quad j) + (\partial\phi/\partial y \quad k) \quad W5 = \quad \nabla\phi \quad W5 \\ & +(\partial A_z/\partial z \quad -l) + (\partial A_x/\partial x \quad -l) + (\partial A_y/\partial y \quad -l) \quad W9 = \quad -\nabla\cdot\mathbf{A} \quad W9 \\ & +(\partial A_z/c\partial t \quad i) + (\partial A_x/c\partial t \quad j) + (\partial A_y/c\partial t \quad k) \quad WD = \quad \partial\mathbf{A}/c\partial t \quad WD \\ & +(\partial A_x/\partial z - \partial A_z/\partial x) k + (\partial A_y/\partial x - \partial A_x/\partial y) i + (\partial A_z/\partial y - \partial A_y/\partial z) j \quad WA = \\ & \nabla\times\mathbf{A} \quad WA \end{aligned}$$

Each derivative is the magnitude of a single sequency.

$$\begin{aligned} \nabla\times\mathbf{A} &= \mathbf{B} \\ -\nabla\phi - \partial\mathbf{A}/c\partial t &= \mathbf{E} \\ \nabla\cdot\mathbf{A} &= 0, \text{ Coulomb guage} \\ \nabla\cdot\mathbf{A} + 1/c^2 \partial\phi/\partial t &= 0, \text{ Lorenz guage} \end{aligned}$$

The rest of Maxwell's Equations follow naturally from the vector identities associated with the operators. Electric and magnetic fields are projections of the same entity. Of course, they are connected, even if they are orthogonal.